

STATISTICAL ORBIT DETERMINATION THEORY

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Abstract. Recent advances in methods for estimating the state of a continuous non-linear dynamic system using discrete observations are reviewed. Particular attention is given to a comparison of batch and sequential algorithms. The stability and convergence characteristics as well as questions related to the computation effort are considered. Finally, numerical results obtained in applying the recently developed Dynamic Model Compensation algorithm are compared with results obtained using a batch algorithm using range-rate data obtained during the Apollo 10 and 11 lunar orbit missions. These results indicate a distinct improvement in estimation accuracy using the Dynamic Model Compensation algorithm.

1. Introduction

The problem of estimating the state of any nonlinear dynamical system influenced by random forces using discrete observations which are subject to random error has received considerable attention during the last decade. The conventional solution to the problem involves linearizing the nonlinear equations about a reference solution and, then, applying linear estimation techniques. The estimate of the deviation from the reference solution can be obtained, then, by either a batch data processing algorithm in which a large batch of observations are used to obtain the estimate of the state at some reference epoch or the state may be estimated sequentially by processing each observation as it is made.

Errors of four basic types influence the accuracy of the linear estimates:

- (1) errors due to the linearization assumptions;
- (2) errors introduced in the computational procedures;
- (3) errors which occur in the observation process; and
- (4) errors due to inaccuracies in the mathematical model used to describe the dynamical process.

While the errors which occur in the observation process are important in determining the overall accuracy of any estimation procedure, they will not be considered in this discussion. Instead, attention will be confined to the effects of the errors encountered in the computation and modeling procedures. In complex orbit determination programs, the errors which occur in the computational process and the errors due to inaccurate modeling are closely related and careful consideration of the errors is required to distinguish the particular cause. With a complex mathematical model, it is difficult to obtain an accurate numerical solution to the relations which define the estimation procedure. On the other hand, if the mathematical model is simplified to alleviate computational problems, important physical effects may be neglected and the estimation procedure may diverge.

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There have been a comparatively few studies related to the problem of orbit determination in the presence of unmodeled dynamic errors (as contrasted with a considerable bibliography of literature which considers the effects of the first three error sources). Since the unmodeled accelerations will vary, generally, with time, the effect of the unmodeled accelerations can be compensated for most easily by using a sequential or Kalman-Bucy type of filter. Furthermore, if the extended form of the sequential estimation procedure is used, where the estimate at a time, t_i , is used to define the reference trajectory for propagating the estimate to the next observation epoch, t_j , then the errors due to the linearization assumptions will be minimized.

These concepts along with several other pertinent points are discussed in the subsequent sections. In Section 2, the general formulation of the orbit determination problem for a near-Earth satellite is discussed, and the problem is reduced to a linear state estimation problem. In Section 3, the 'batch' minimum variance (or 'least squares') solution to the problem is discussed. In Section 4, an alternate procedure for solving the normal equations associated with the batch estimate is described. In Section 5, the sequential estimation algorithm (or Kalman-Bucy filter) is derived directly from the minimum-variance batch algorithm. In Section 6, the problem of estimating the state in the presence of unmodeled accelerations is discussed and the application of the Dynamic Model Compensation algorithm to the lunar orbit determination problem is described. In Section 7, a summary of the conclusions discussed in the previous sections is given.

2. Problem Formulation

The equations which govern the motion of a vehicle moving in a central force field can be expressed as

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \frac{\mu \mathbf{r}}{r^3} + \mathbf{R}(\mathbf{r}, t) \quad (2.1)$$

where $\mathbf{r}$ is the position vector, $\mathbf{v}$ is the velocity vector, μ is the gravitational parameter, and $\mathbf{R}$ is the vector of perturbations. Equations (2.1) can be expressed in first order form as follows.

$$\begin{aligned} \dot{\boldsymbol{\zeta}} &= \mathbf{f}(\boldsymbol{\zeta}, \boldsymbol{\alpha}, t) \\ \dot{\boldsymbol{\alpha}} &= \mathbf{0} \end{aligned} \quad (2.2)$$

where $\boldsymbol{\alpha}$ is a q -vector of all constant parameters in Equations (2.1) whose values are to be estimated during the orbit determination procedure. The six-vectors $\boldsymbol{\zeta}$ and $\mathbf{f}(\boldsymbol{\zeta}, \boldsymbol{\alpha}, t)$ are defined as

$$\boldsymbol{\zeta}^T = [\mathbf{r}^T : \mathbf{v}^T] \quad \mathbf{f}^T(\boldsymbol{\zeta}, \boldsymbol{\alpha}, t) = [\mathbf{v}^T : (-\mu \mathbf{r}/r^3 + \mathbf{R})^T]. \quad (2.3)$$

A. THE STATE EQUATIONS

As defined, the state vector is composed of all dependent variables or constant parameters required to define the time rate of change of the state of the dynamical system. With this definition, the n -dimensional state vector, $\mathbf{X}$, can be expressed as follows